WHATNEXT VISION MOTORS

Salesforce CRM - Vehicle Order Management System

PHASE 5 - PROJECT DEVELOPMENT PHASE

Dedicated Phase Documentation

Development Scope	Data Modelling, Fields and Relationships, Lightning App Builder, Record-Triggered Flows, Apex, Batch Apex and Scheduled Apex
Source	Prepared from the uploaded WhatNext Vision Motors project documentation

This document separates the complete Project Development Phase into its implementation milestones for project review and submission.

PHASE 5 - PROJECT DEVELOPMENT PHASE

5.1 Learning / Development Areas

The development phase converts the project design into an implemented Salesforce CRM solution. The work progresses from Salesforce environment setup through data modelling, user interface configuration, automation, programmatic business logic, batch processing and scheduled execution.

Development Flow	Salesforce Account -> Objects -> Tabs -> Lightning App -> Fields & Relationships -> Record-Triggered Flows -> Apex Trigger -> Batch Apex -> Scheduled Apex

Development Objectives

- Create and activate the Salesforce Developer environment.
- Build the core data model for vehicles, dealers, customers, orders, test drives and service requests.
- Provide custom tabs and a centralized Lightning application.
- Define fields and lookup relationships required for data capture and automation.
- Automate dealer assignment and test-drive reminders with Record-Triggered Flows.
- Use Apex to validate stock and update stock for confirmed orders.
- Process pending orders with Batch Apex and automate execution with Scheduled Apex.

5.2 Milestone 1 - Salesforce Account

Purpose

Create and activate a Salesforce Developer account that provides the environment required to configure the WhatNext Vision Motors CRM solution.

Activity 1: Creating Developer Account

- Open the Salesforce Developer signup page.
- Enter first name and last name.
- Enter the email address.
- Select the role as Developer.
- Enter the college or company name.
- Select India as the country.
- Enter the postal code.
- Create a username and submit the signup form.

Activity 2: Account Activation

- Open the email inbox used during signup.
- Click the account verification link.
- Set a password and answer the security question.
- Complete activation and access the Salesforce Setup page.

Expected Outcome

A working Salesforce Developer account is available and the project team can access Salesforce Setup for further configuration.

5.3 Milestone 2 - Data Management: Objects

Purpose

Create the custom objects that represent the main entities in the vehicle ordering and dealership process.

Object	Purpose	Relationships
Vehicle__c	Stores vehicle details	Dealer and Orders
Vehicle_Dealer__c	Stores authorized dealer information	Orders
Vehicle_Customer__c	Stores customer details	Orders and Test Drives
Vehicle_Order__c	Tracks vehicle purchases	Customer and Vehicle
Vehicle_Test_Drive__c	Tracks test-drive bookings	Customer and Vehicle
Vehicle_Service_Request__c	Tracks vehicle servicing requests	Customer and Vehicle

Activity: Create Vehicle Object

- Open Setup -> Object Manager -> Create -> Custom Object.
- Label: Vehicle; Plural Label: Vehicles.
- Record Name: Vehicle Name; Data Type: Text.
- Enable Allow Reports and Allow Search.
- Save the object.

Activity: Create Dealer Object

- Open Object Manager -> Create -> Custom Object.
- Label: Vehicle Dealer; Plural Label: Vehicle Dealers.
- Record Name: Dealer Name; Data Type: Text.
- Enable Allow Reports and Allow Search.
- Save the object.

Remaining Objects

Create the Customer, Order, Test Drive and Service Request objects using the same custom-object creation process and configure their relationships according to the project model.

5.4 Milestone 3 - Data Management: Tabs

Purpose

Expose the custom objects through Salesforce navigation so users can access project records easily.

- Open Setup -> Tabs -> New under Custom Object Tabs.
- Select the Vehicle object.
- Choose an appropriate tab style.
- Retain the required profile and app settings.
- Save the tab.
- Repeat the process for the remaining project objects.

Expected Outcome

Custom tabs are available for the project objects and can be added to the Lightning application.

5.5 Milestone 4 - Data Management: App Manager

Purpose

Create a centralized Lightning application named WhatNext Vision Motors and place the important project components in its navigation.

- Open App Manager and create a new Lightning App.
- Set the application name to WhatNext Vision Motors.
- Add Vehicle, Dealer, Customer, Order, Test Drive, Service Request, Reports and Dashboard as navigation items.
- Add the System Administrator profile.
- Save the application.

Expected Outcome

Users can access project data and reporting components from a single Salesforce Lightning application.

5.6 Milestone 5 - Data Management: Fields and Relationships

Purpose

Define the data fields required to capture project information and connect related records through lookup relationships.

Object	Key Fields
Vehicle__c	Vehicle_Name__c (Text); Vehicle_Model__c (Picklist); Stock_Quantity__c (Number); Price__c (Currency); Dealer__c (Lookup); Status__c (Picklist)
Vehicle_Dealer__c	Dealer_Name__c (Text); Dealer_Location__c (Text); Dealer_Code__c (Auto Number); Phone__c (Phone); Email__c (Email)
Vehicle_Order__c	Customer__c (Lookup); Vehicle__c (Lookup); Order_Date__c (Date); Status__c (Picklist)
Vehicle_Customer__c	Customer_Name__c (Text); Email__c (Email); Phone__c (Phone); Address__c (Text); Preferred_Vehicle_Type__c (Picklist)
Vehicle_Test_Drive__c	Customer__c (Lookup); Vehicle__c (Lookup); Test_Drive_Date__c (Date); Status__c (Picklist)
Vehicle_Service_Request__c	Customer__c (Lookup); Vehicle__c (Lookup); Service_Date__c (Date); Issue_Description__c (Text); Status__c (Picklist)

Field Creation

- Open Setup -> Object Manager -> required object -> Fields & Relationships -> New.
- Select the required data type.
- Enter the field label and configure picklist values or other field settings.
- Proceed through the configuration and save the field.

Lookup Relationship

Create a Lookup Relationship field in the Vehicle object and select the Dealer object as the related object. This allows a vehicle record to reference its associated dealer.

5.7 Milestone 6 - Automation: Auto Assign Dealer Flow

Objective

Create a Record-Triggered Flow that assigns a dealer to a vehicle order based on customer location information.

Flow Configuration

- Open Flows -> New Flow -> Start From Scratch -> Record-Triggered Flow.
- Select Vehicle Order as the object.
- Trigger when a record is created.
- Entry condition: Status__c Equals Pending.
- Get Records: retrieve customer information using the customer ID.
- Get Records: identify the nearest dealer using the customer address or location.
- Update Records: assign the dealer to the order.
- Save the flow as Auto Assign Dealer and activate it.

Logical Flow

New Pending Order -> Get Customer Information -> Find Nearest Dealer -> Assign Dealer -> Continue Order Processing

5.8 Milestone 6 - Automation: Test Drive Reminder Flow

Objective

Create a Record-Triggered Flow that automatically sends a reminder email before a scheduled test drive.

- Create a Record-Triggered Flow for Vehicle Test Drive.
- Trigger when a record is created or updated.
- Entry condition: Status__c Equals Scheduled.
- Add a scheduled path one day before Test_Drive_Date__c.
- Get the customer information.
- Add a Send Email action.
- Subject: Reminder: Your Test Drive is Tomorrow!
- Use the customer Email__c as the recipient.
- Save and activate the flow.

Expected Outcome

A scheduled test drive generates an automated reminder to the customer one day before the test-drive date.

5.9 Milestone 7 - Apex Trigger Handler

Purpose

VehicleOrderTriggerHandler centralizes Apex logic for vehicle orders. Before operations validate stock availability; after operations update stock when a confirmed order is processed.

Stock Validation

The handler retrieves vehicle stock for the vehicles associated with incoming orders. When Stock_Quantity__c is less than or equal to zero, the order is prevented and the project uses the message below.

```
This vehicle is out of stock. Order cannot be placed.
```

Stock Update

When an order has Status__c = Confirmed, the corresponding vehicle stock is reduced by one and the updated vehicle record is saved.

Apex Trigger

The Vehicle Order trigger is configured for before insert, before update, after insert and after update. It calls VehicleOrderTriggerHandler.handleTrigger with the relevant trigger context values.

Trigger Handler Structure

```
public class VehicleOrderTriggerHandler {
public static void handleTrigger(List<Vehicle_Order__c> newOrders,
Map<Id, Vehicle_Order__c> oldOrders, Boolean isBefore,
Boolean isAfter, Boolean isInsert, Boolean isUpdate) {
if (isBefore && (isInsert || isUpdate)) {
preventOrderIfOutOfStock(newOrders);
}
if (isAfter && (isInsert || isUpdate)) {
updateStockOnOrderPlacement(newOrders);
}
}
}
```

Expected Outcome

The trigger prevents invalid out-of-stock orders and updates vehicle stock for eligible confirmed orders.

5.10 Milestone 7 - Batch Apex

Purpose

Process pending vehicle orders in manageable batches and confirm eligible orders when stock becomes available.

- Identify pending vehicle orders that can be processed.
- Process records in batches instead of attempting to handle all pending orders at once.
- Check current vehicle stock availability for eligible pending orders.
- Change eligible pending orders to Confirmed when stock is available.
- Reduce the corresponding vehicle stock by one for each confirmed order.
- Save the updated records.

Processing Logic

Pending Orders -> Batch Query -> Stock Check -> Confirm Eligible Orders -> Reduce Stock -> Save Updates

Expected Outcome

Eligible pending orders are processed automatically and vehicle stock is synchronized with confirmed orders.

5.11 Scheduled Apex

Purpose

Automate execution of the batch process so pending orders can be processed on a schedule without manual execution.

- Create the VehicleOrderBatchScheduler Apex class.
- Configure the scheduler to execute VehicleOrderBatch.
- Use a batch size of 50 for batch execution.
- Schedule the Apex job to run automatically at the required interval.
- Monitor the scheduled job from Salesforce Setup when required.

Expected Outcome

VehicleOrderBatch is launched automatically through the scheduled Apex process, supporting continuous processing of eligible pending orders.

PHASE 5 - DEVELOPMENT SUMMARY

Milestone Summary

Milestone	Main Deliverable	Primary Salesforce Feature
1	Developer environment	Salesforce Developer Account
2	Core data model	Custom Objects
3	Object navigation	Custom Tabs
4	Central application	Lightning App Manager
5	Data capture and relationships	Fields and Lookup Relationships
6A	Automatic dealer assignment	Record-Triggered Flow
6B	Test-drive reminder	Scheduled Path + Send Email Flow
7A	Order validation and stock update	Apex Trigger + Handler
7B	Pending-order processing	Batch Apex
7C	Automatic batch execution	Scheduled Apex

Development Phase Outcome

The Project Development Phase establishes the working Salesforce implementation for WhatNext Vision Motors. It combines structured data management, a centralized Lightning interface, declarative automation, programmatic business logic, batch processing and scheduled execution to support the vehicle ordering workflow.

End-to-End Development Flow

Salesforce Account -> Objects -> Tabs -> Lightning App -> Fields & Relationships -> Auto Assign Dealer Flow -> Test Drive Reminder Flow -> Apex Trigger Handler -> Batch Apex -> Scheduled Apex

Source note: Content is based on the uploaded WhatNext Vision Motors project documentation, reorganized as a dedicated Phase 5 document.